

Minute-02

Date: 2026-06-15

PROJECT NAME: Plagiarism Detector

SUPERVISOR: Er. Prativa Nyaupane

TEAM LEADER (TL): Neon Neupane

MEMBER(M1): Bishal Acharya

MEMBER(M2): Nabin Giri

PROGRESS CHECK: (Tick ✓ in appropriate box)

☐ 5 ☐ 10 ☐ 15 ☐ 20 ☐ 25 ☐ 30 ☐ 35 ☐ 40 ☐ 45 ☐ 50 ☐ 55 ☐ 60
☐ 65 ☐ 70 ☐ 75 ☐ 80 ☐ 85 ☐ 90 ☐ 95 ☐ 100

AGENDA:

- Presentation of the review of the text similarity techniques
- Finalization of the dataset and of the model
- Freezing of the technology stack and of the repository structure
- Review of the plan document and of the pipeline document

DISCUSSION:

The team leader presented the review of the similarity techniques, which covered three families, that is, exact string and n-gram matching, TF-IDF with cosine similarity, and transformer based sentence embeddings. The first two cannot detect a paraphrase, so sentence embeddings were selected as the main method. The three members compared the available labelled corpora together and selected the Quora Question Pairs corpus for training, because it carries about four hundred thousand labelled pairs and can be downloaded without an account, and Sentence-BERT with the all-MiniLM-L6-v2 base was selected as the model, to be fine tuned with a Siamese architecture in which twin networks that share the same weights turn two sentences into vectors and the training pulls similar pairs closer. The lightweight MiniLM variant was chosen because it runs on the CPU of the laptops available to the team. The stack was frozen as Django with the REST framework for the backend, React with Vite and Tailwind CSS for the frontend, PyTorch for the model and Tesseract for OCR. The supervisor asked the team to keep TF-IDF as a baseline so that the report can show a measurable comparison, and to record the accuracy of both engines during testing. The repository was created in this week and the plan document, which carries the objectives, the dataset references and the module wise pipeline, was committed to it.

INDIVIDUAL ACHIEVEMENT (LAST SPRINT):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>
Created the repository and committed the plan document along with the document that describes the processing pipeline and the responsibility of each module, tested the PDF and DOCX reading libraries and Tesseract OCR on real files, and recorded which library handled scanned and two column pages best.	Reviewed the research papers and articles on semantic similarity, compared exact string matching, TF-IDF with cosine similarity and transformer based sentence embeddings, and selected Sentence-BERT with the all-MiniLM-L6-v2 base as the model to be fine tuned, the lightweight variant being chosen because it runs on the processor of the laptops available to the team.	Learned React and Vite, prepared the wireframes of the upload, matrix and result screens, and presented the complete user flow of the application to the team before the layout was fixed.

RESPONSIBILITIES (COMING SPRINT):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>
Prepare the data flow diagrams, the use case diagram and the sequence diagram, and fix the request and response format of the API so that the two sides can be built in parallel.	Prepare the training data loader and the evaluation script, and run a trial fine tune on a small sample of the corpus to confirm that the pipeline works before the full run.	Convert the wireframes into a component list for React and fix the Tailwind theme, that is, the colours, the typography and the page layout.

SIGNATURE:

MEMBER 1

MEMBER 2

TEAM LEADER

SUPERVISOR

To be filled by Supervisor:

INDIVIDUAL MARKING (5):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>